

SOUJANYA S P

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SUMMARY

AI Engineer with hands-on experience building agentic AI systems, LLM-based workflow automation, and production-ready backend services using Python, FastAPI, LangChain, and LangGraph. Skilled in deploying scalable systems with Docker and AWS (EC2, S3, Lambda), with a strong foundation in REST API development, multi-agent orchestration, and CI/CD.

PROJECTS

AI Voice Assistant — VAPI · FastAPI · Pandas · SQLite · Streamlit · Agentic Architecture [Link](#)

- ▶ Built an end-to-end agentic voice assistant using VAPI and FastAPI that converts unstructured speech inputs into structured database records, automating hospital appointment management and reducing manual data entry by ~70%.
- ▶ Implemented end-to-end workflow using Pandas and SQLite to ingest patient data and produce Excel insights, eliminating manual reporting and saving approximately ~3 hours/week.

Self-Validating Quant Word-Problem Generator - Python · Multi-Agent AI · LLM Orchestration [Link](#)

- ▶ Architected a production-style multi-agent AI system comprising research, problem-generation, dual-solver, validation, and orchestration agents to autonomously generate and verify quantitative word problems, eliminating manual review across generated quiz sets.
- ▶ Integrated hallucination detection and mathematical correctness verification mechanisms, enabling accelerated validation of generated content and producing accuracy reports with zero manual intervention.

AI Research & Report Generation System - LangGraph · FastAPI · Streamlit · Tavily · Agentic AI [Link](#)

- ▶ Developed an autonomous multi-agent AI research platform using LangGraph and LangChain, orchestrating six specialized agents for planning, web research, content generation, fact-checking, and report assembly.
- ▶ Orchestrated an end-to-end research workflow integrating Tavily Search, LLM-powered summarization, credibility validation, and citation tracking to generate structured academic-style reports with AI-driven trust scoring.

EXPERIENCE

AI Developer Intern | Workfall

May 2026 - Aug 2026 · Bangalore

- ▶ Enabled AI agent workflows capable of interpreting natural language instructions and automating task creation, assignment, and workflow management within a collaborative work management platform.
- ▶ Created LLM-powered automation pipelines using Python and API integrations, enabling intelligent task orchestration, workflow automation, and streamlined execution.

Backend Engineer Intern | Sanskriti Labs

Oct 2025 – Dec 2025 · Bangalore

- ▶ Established RESTful API services using NestJS with JWT authentication, middleware handling, and TypeORM for PostgreSQL integration.
- ▶ Optimized ETL-style data integration pipelines, improving data flow efficiency by ~40% across structured and unstructured data sources.

SKILLS

- ▶ **Programming Languages:** Python, SQL
- ▶ **AI & LLM Development:** LLMs, Prompt Engineering, RAG, Embeddings, Vector Search, AI Agents, Multi-Agent Systems, Function Calling, Structured Outputs, LLM Evaluation, AI Workflow Orchestration, MCP, LangChain, LangGraph
- ▶ **Frameworks & Tools:** FastAPI, Streamlit
- ▶ **Databases & Vector Stores:** SQLite, ChromaDB, FAISS
- ▶ **Software Engineering:** REST APIs, API Integration, JSON, Workflow Automation, Debugging & Testing
- ▶ **Cloud & Deployment:** AWS (EC2, S3, Lambda), Docker, Git, GitHub, CI/CD, Deployment Automation

EDUCATION

MS RAMAIAH COLLEGE OF ARTS, SCIENCE, AND COMMERCE – Bangalore

2024 – 2026

Master of Computer Applications (MCA) | Bengaluru City University

PES INSTITUTE OF ADVANCED MANAGEMENT STUDIES, Shivamogga

2021 – 2024

Bachelor of Computer Applications (BCA) | Kuvempu University

CGPA: 9.38 / 10

MAHESH PU COLLEGE, Shivamogga

2020 – 2021

Percentage: 98%